STUDENT REPORT

DETAILS

Name

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Roll Number

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EXPERIMENT

Title

WEEK 9A

Description

9 .Write R program to find the following Regressions with the sample data and visualize the regressions graphically.

a) Linear Regression

Procedure and Overvation:

Regression analysis is a very widely used statistical tool to establish a relationship model between two variables. One of these variable is called predictor va riable whose value is gathered through experiments. The other variable is called response variable whose value is derived from the predictor variable.
DESCRIPTION:
In Linear Regression these two variables are related through an equation, where exponent (power) of both these variables is 1. Mathematically a linear relat ionship represents a straight line when plotted as a graph. A non-linear relationship where the exponent of any variable is not equal to 1 creates a curve.

The general mathematical equation for a linear regression is −

y = ax + b

Following is the description of the parameters used −

y is the response variable.

x is the predictor variable.

a and b are constants which are called the coefficients.

Steps to Establish a Regression

A simple example of regression is predicting weight of a person when his height is known. To do this we need to have the relationship between height and weight of a person.

The steps to create the relationship is −

1)Carry out the experiment of gathering a sample of observed values of height and corresponding weight.

2)Create a relationship model using the lm() functions in R.

3)Find the coefficients from the model created and create the mathematical equation using these

4)Get a summary of the relationship model to know the average error in prediction. Also called residuals.

5)To predict the weight of new persons, use the predict() function in R.

lm() Function
This function creates the relationship model between the predictor and the response variable.

Syntax
The basic syntax for lm() function in linear regression is −

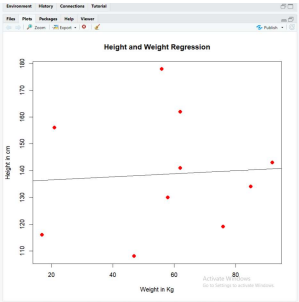
lm(formula,data)
Following is the description of the parameters used −

formula is a symbol presenting the relation between x and y.

data is the vector on which the formula will be applied.
predict()function
predict(object , newdata)
Object is the formula which is already created using the lm() function.
Newdata is the vector containing the new value for predictor variable
SOURCE CODE

```
# Create the predictor and response variable.
x <- c(141,134,178,156,108,116,119,143,162,130)
y <- c(62,85,56,21,47,17,76,92,62,58)
relationship_model<-lm(y~x)
# Give the chart file a name.
png(file = "linear_regression.png")
# Plot the chart.
plot(y,x,col = "red",main = "Height and Weight Regression",
    abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in cm")
# Save the file.
```

```
dev.off()
```



VIVA RESPONSES

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